

Multiple Choice (10 marks)

1. What is the dimensionality of the null space of the following matrix? $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
 - (a) 0
 - (b) 1
 - (c) 2
 - (d) 3
2. As the number of training examples goes to infinity, your model trained on that data will have:
 - (a) Lower variance
 - (b) Higher variance
 - (c) Same variance
 - (d) None of the above
3. The disadvantage of Grid search is
 - (a) It can not be applied to non-differentiable functions.
 - (b) It can not be applied to non-continuous functions.
 - (c) It is hard to implement.
 - (d) It runs reasonably slow for multiple linear regression.
4. The number of test examples needed to get statistically significant results should be __
 - (a) Larger if the error rate is larger.
 - (b) Larger if the error rate is smaller.
 - (c) Smaller if the error rate is smaller.
 - (d) It does not matter.
5. Statement 1| ImageNet has images of various resolutions. Statement 2| Caltech-101 has more images than ImageNet.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: Underfitting occurs when a model fails to capture the underlying patterns in the training data, leading to poor performance on both training and new data.
7. True or False: Linear soft-margin SVM can only be used with linearly separable training data.
8. True or False: Batch Normalization is used in the original ResNet paper, while Layer Normalization is not.
9. True or False: The use of sampling with replacement is the primary factor that prevents overfitting when performing bagging.
10. True or False: To calculate $P(H|E, F)$, the sufficient numbers include $P(E, F)$, $P(H)$, and $P(E, F|H)$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to count the number of inversions in the given array.
12. Write a python function to check whether the sum of divisors are same or not.

End of Paper